
mechanical checks, where all soundness lives
given
query (A, G) in PolySTL
architecture $M_1 \to M_2$
code of M_2, draft of M_1
proposer agents
read the code,
propose (A_1, G_1)
and (A_2, G_2)
phase 1, composition
D1 $A \subseteq A_1$ D2 $G_1 \subseteq A_2$
D3 $A \cap G_1 \cap A_2 \cap G_2 \subseteq G$
R $\forall u \in A_1\, \exists w\, G_1(u, w)$
pass
phase 2, discharge
$M_1 \models (A_1, G_1)$, static check
$M_2 \models (A_2, G_2)$, reachability
or bounded model checking, once
pass
verified delegates
$M_1 \cap M_2 \models (A, G)$
by the composition rule
counterexample (u, w, e, y), or an input u with no mission
failing run of M_2, or an input u with $G_1(u, M_1(u))$ false